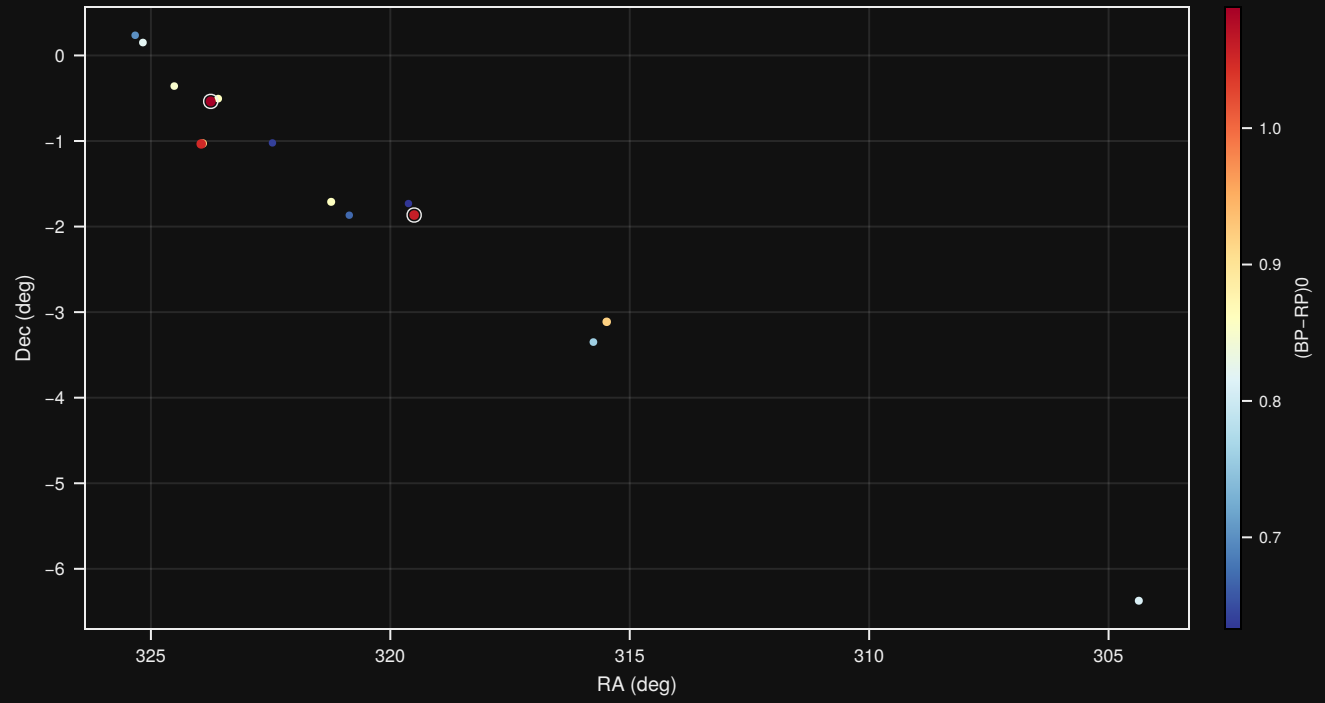
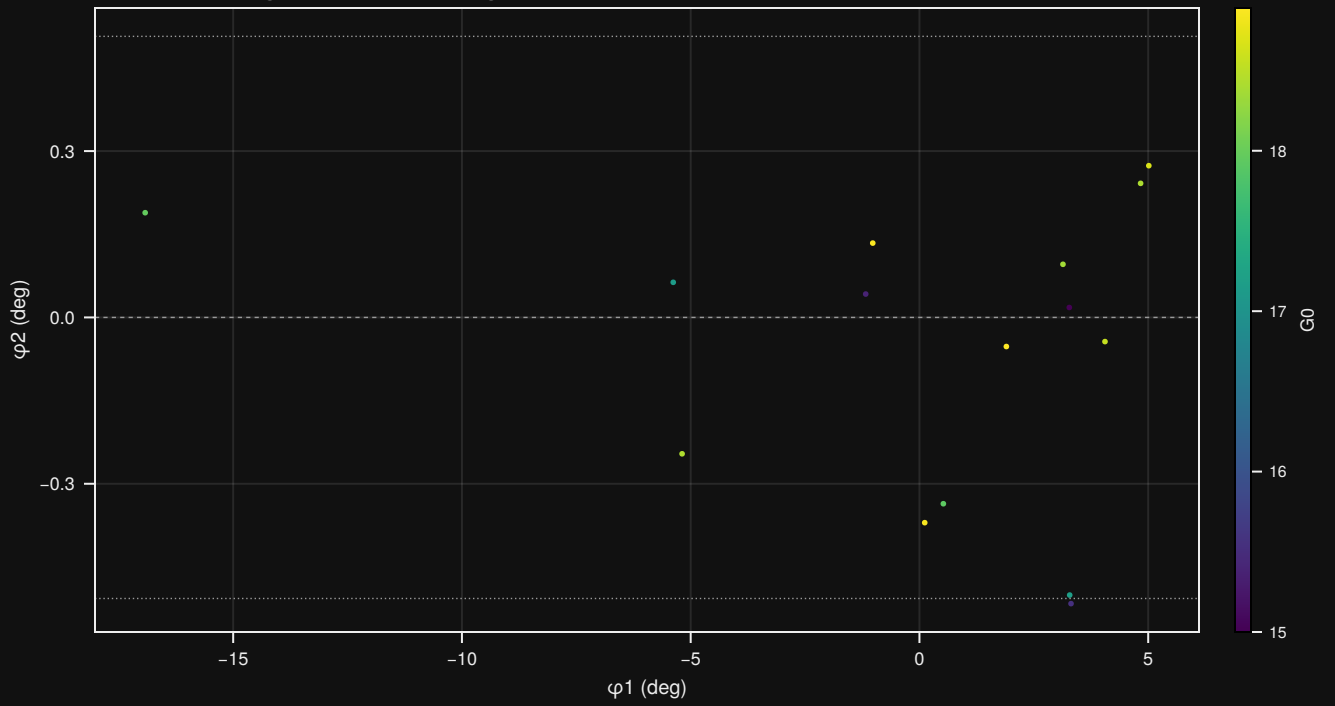
NGC7089

MEMBERSHIP	
Stars in catalog	15
with Gaia GSP-Phot	13 (87%)
with radial velocity	2
Stream id (sID)	76
GEOMETRY · great-circle fit	
Centroid RA, Dec	320.6, -1.0 deg
Centroid l, b	51.0, -35.1 deg
Pole RA, Dec	235.5, 72.3 deg
Length (φ1, 2–98%)	21.9 deg
Width σ(φ2), robust	0.25 deg
KINEMATICS	
pmRA* med ± σ	3.53 ± 0.45 mas/yr
pmDE med ± σ	-2.21 ± 0.07 mas/yr
μφ1* med ± σ	2.74 ± 0.33 mas/yr
μφ2 med ± σ	-3.25 ± 0.18 mas/yr
Parallax (cat.) med	0.079 ± 0.098 mas
Parallax (wtd, ZP-corr.)	0.096 ± 0.020 mas
Distance	d = 10.5 ± 2.2 kpc
Physical length	4.01 kpc
RV median ± σ(MAD)	-1.2 ± 2.3 km/s
RV gradient dV/dφ1	—
PHOTOMETRY · POPULATIONS	
G0 range (median)	15.0–18.9 (18.4)
(BP−RP)0 median	0.85
GSP-Phot [M/H] (n=13)	-1.52 ± 0.71
GSP-Phot Teff median	5397 K
RUWE > 1.4	0.0 %
PM clusters (largest)	none
IBATA+2024 · TABLE 3	
n catalog | paper	15 | 15
n_RV catalog | paper	2 | 2
<[Fe/H]> paper	-1.65
Stellar mass (paper)	300 Msun
Pole offset vs paper	0.6 deg

Sky map · catalog members (n = 15) · size ∝ brightness · ring = has RV



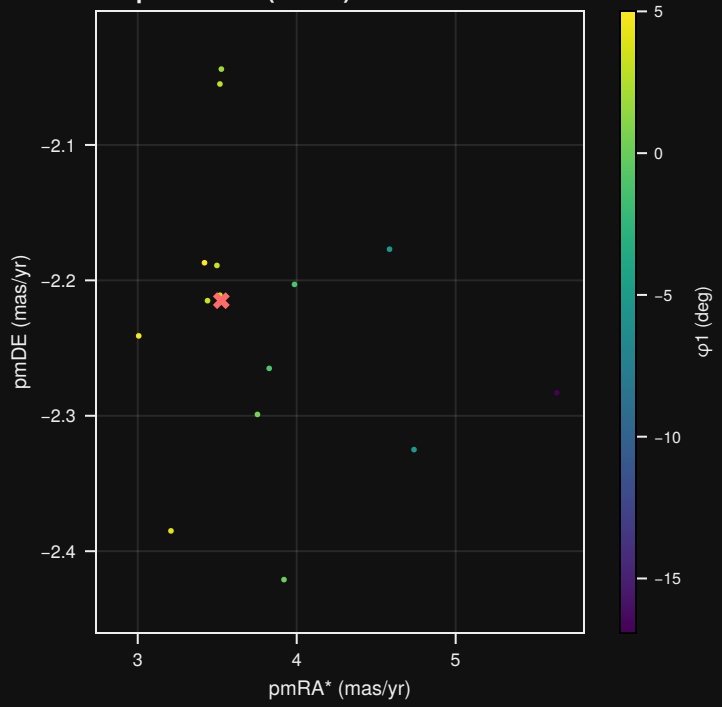
Stream frame (great-circle fit) · length 21.9° · width σ 0.25°



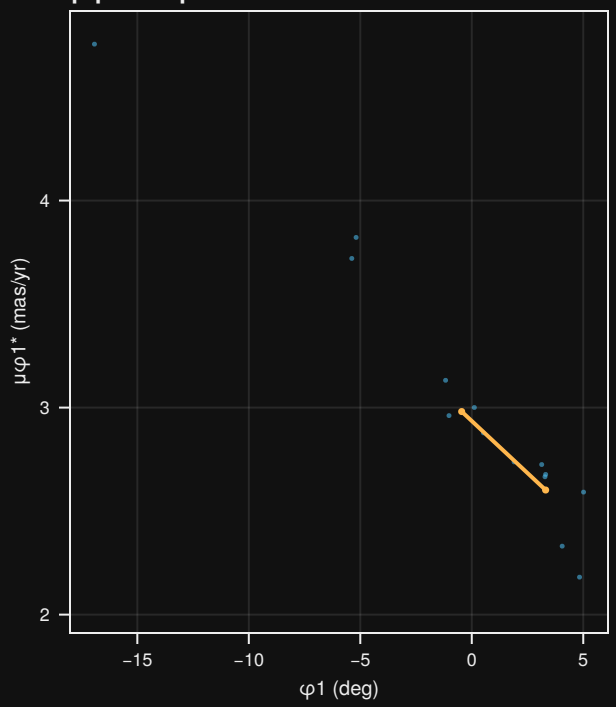
NGC7089

Kinematics · proper motion · parallax · radial velocity

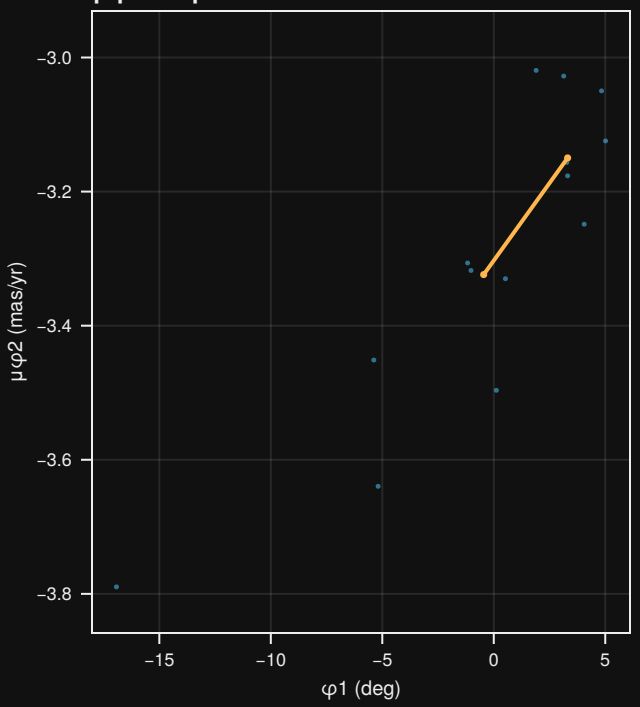
Proper motion (n = 15)



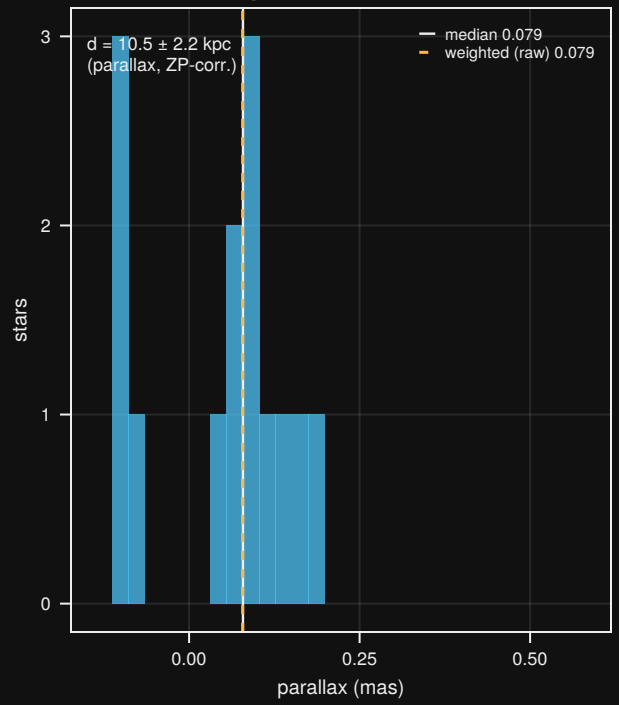
μφ1* vs φ1 · σ = 0.33



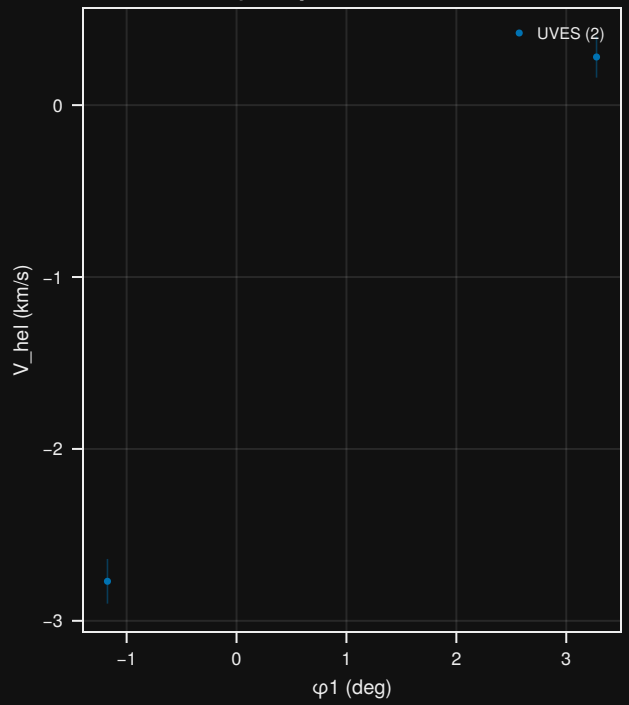
μφ2 vs φ1 · σ = 0.18



Parallax (catalog, n = 15)



Radial velocity vs φ1 (n = 2)

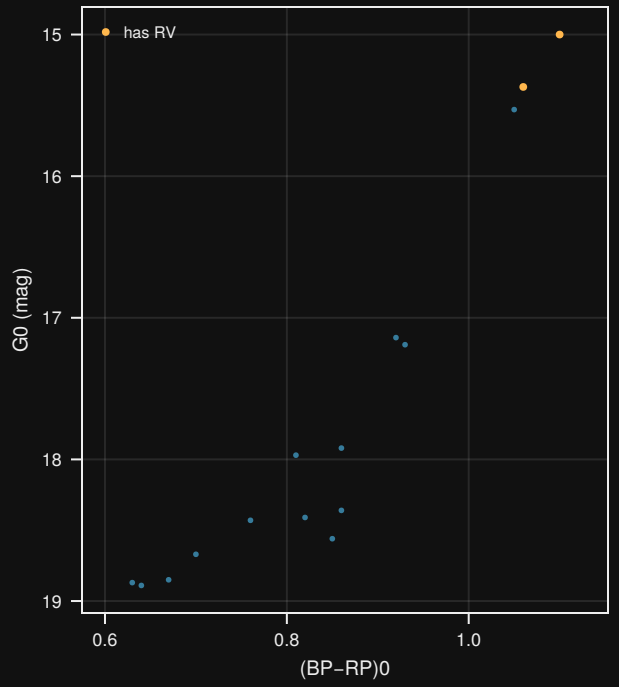


V_hel distribution

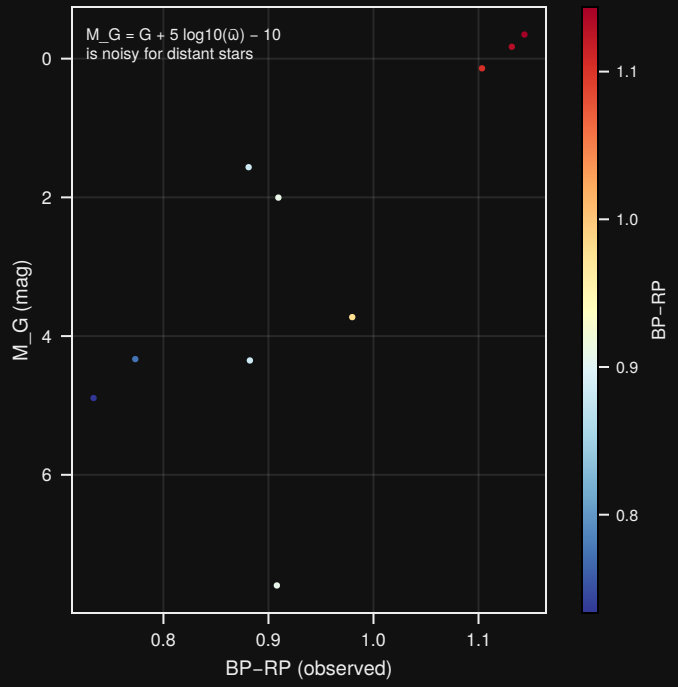
n_RV = 2 (< 3)

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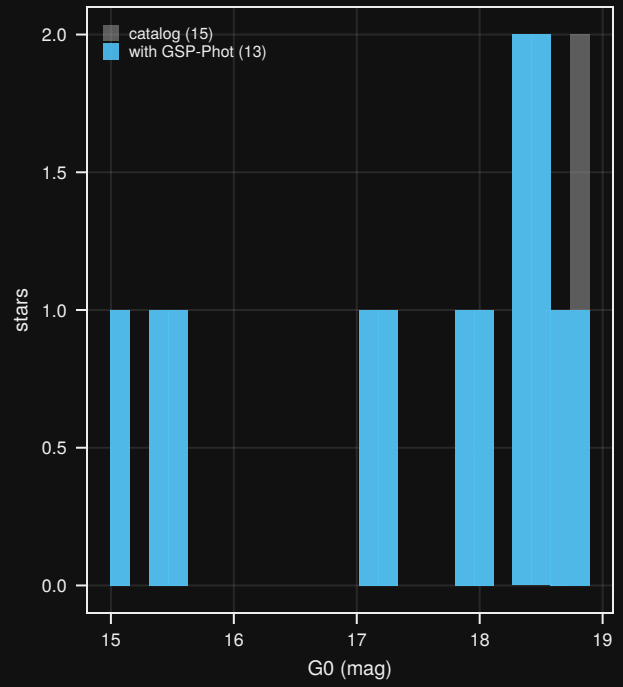
CMD · dereddened (n = 15)



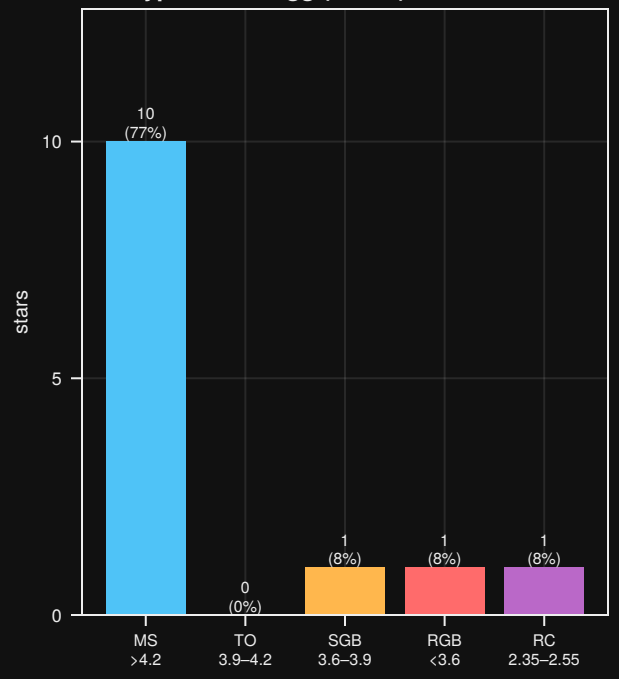
HR diagram · ϖ>0 (n = 10/13)



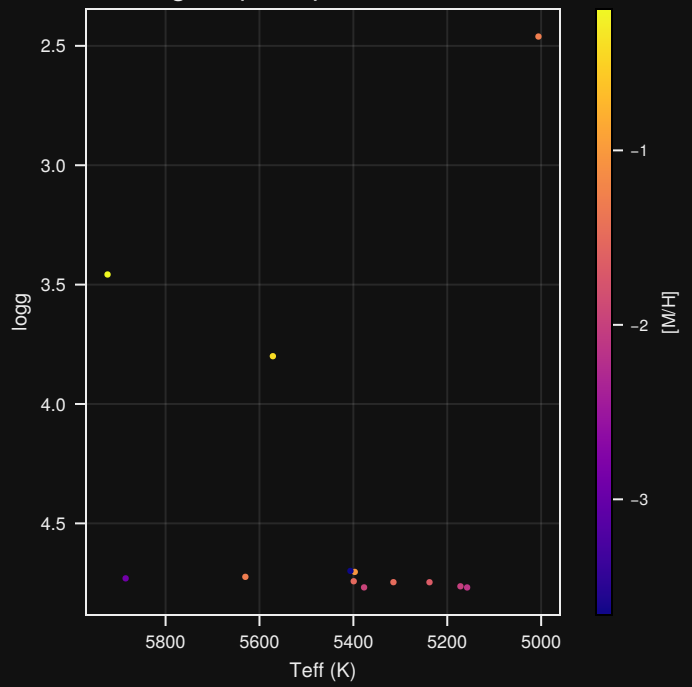
G0 & GSP-Phot coverage



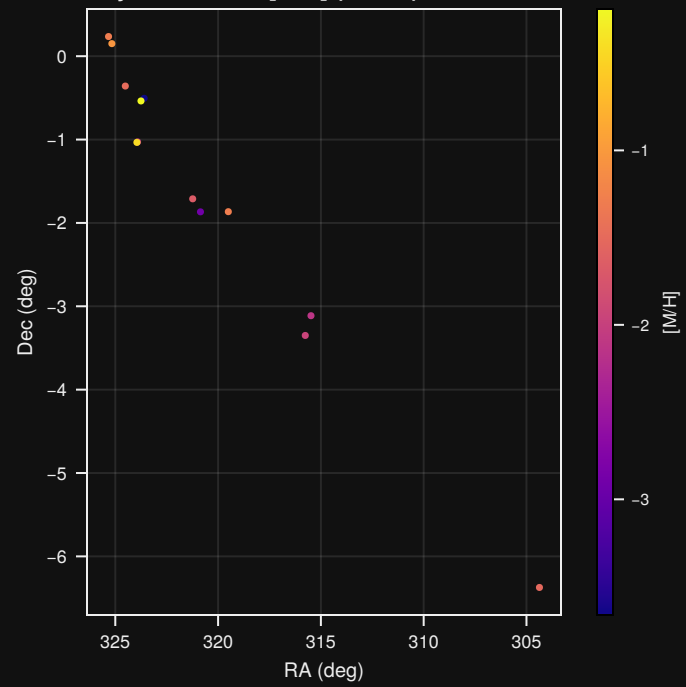
Star types from logg (n = 13)



Kiel diagram (n = 13)



Sky · GSP-Phot [M/H] (n = 13)

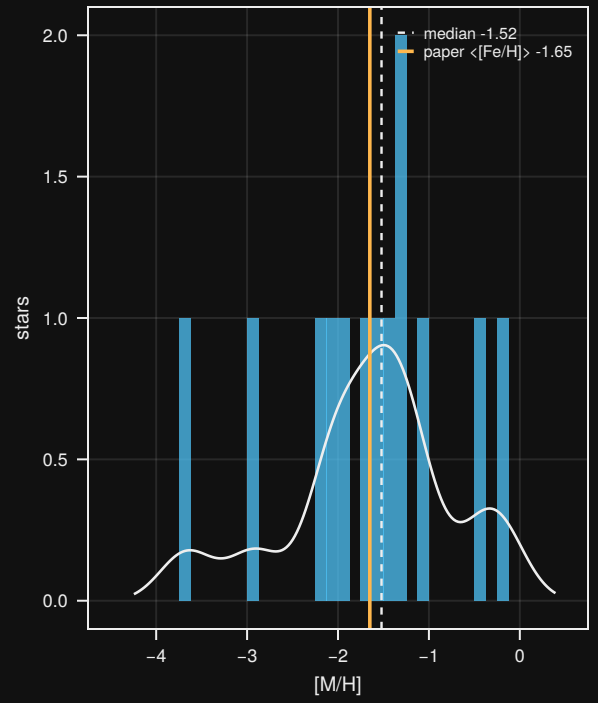


GSP-Phot (Gaia low-resolution BP/RP) parameters are prior-dominated for faint, distant halo stars (G ≥ 17): logg, [M/H] and ages are indicative only and typically biased toward dwarfs.

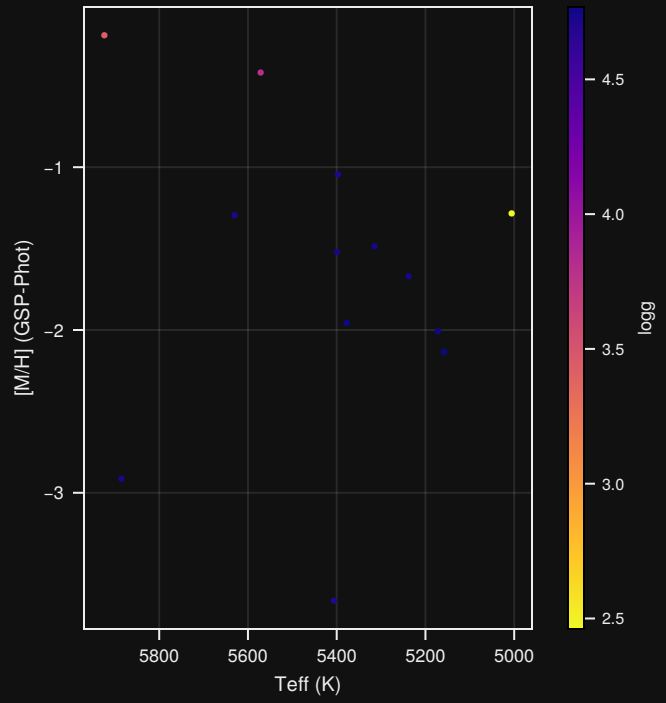
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Metallicity · age · astrometric quality

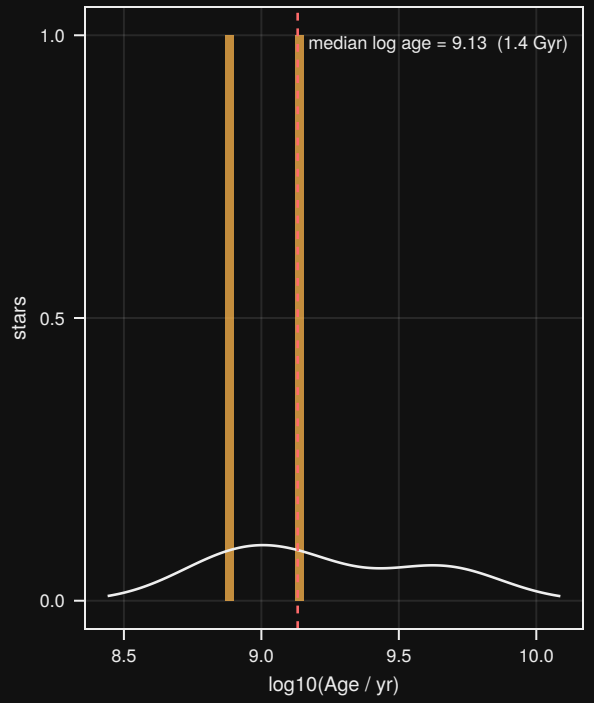
[M/H] GSP-Phot (n = 13)



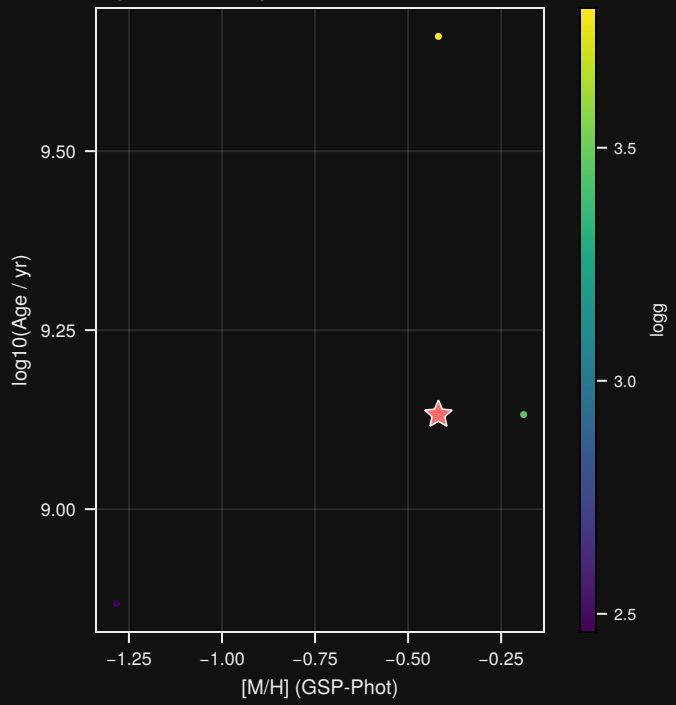
Teff vs [M/H] (n = 13)



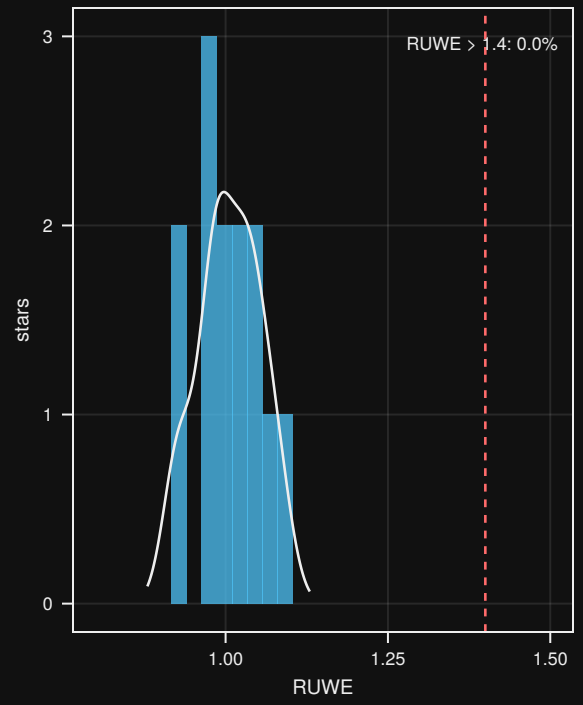
FLAME age (n = 3)



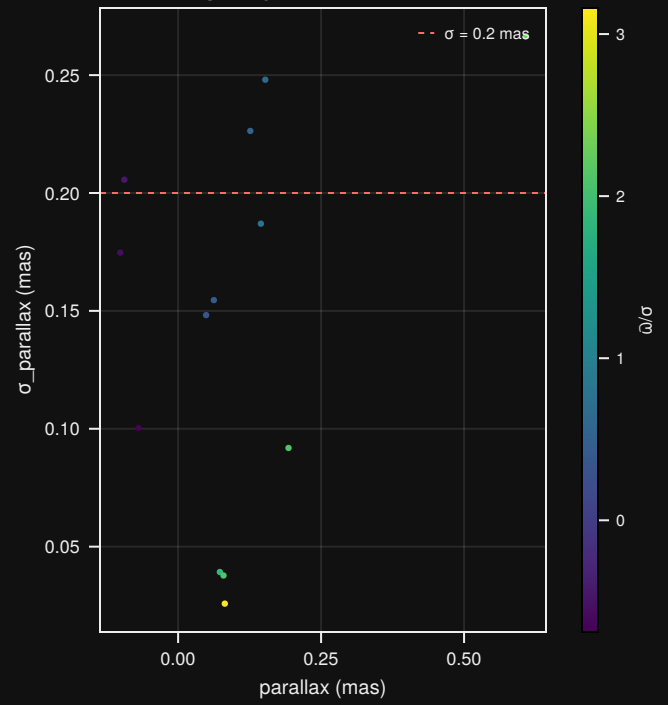
Age–metallicity (n = 3)



RUWE (n = 13)

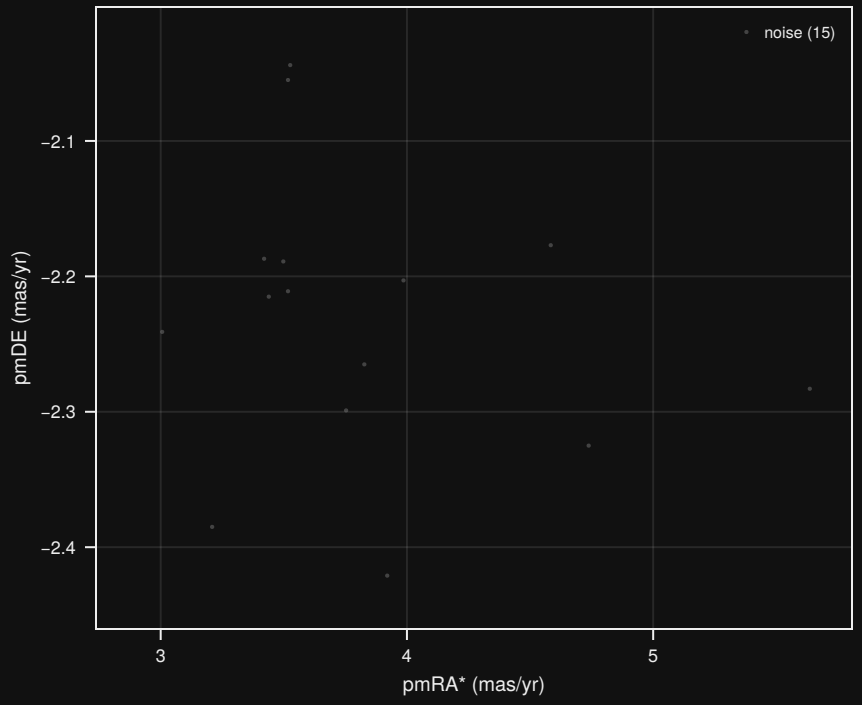


Parallax quality (n = 13)



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PM clustering · HDBSCAN · 0 cluster(s) · noise 100%



Proper-motion directions (arrows) coloured by cluster



Cluster summary

id	n	pmRA	pmDE	σpmRA	[M/H]	n_RV
noise	15	3.53	-2.21	0.45	-1.52	2